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ANNUAL EVALUATION 2024 - 2025

A | **MATHEMATICS EM – ANSWER KEY** | **E-803**

Qn no.	Key	Score	
	Each questions from 1 to 5 carries 2 scores. (Answer any 4)		
1	a) $PQ : QR = 3 : 1$ b) $QR = \frac{1}{3} \times 6 = 2\ cm$	1 1	2
2	a) $\frac{1}{25} = \frac{1 \times 4}{25 \times 4} = \frac{4}{100}$ **OR** $\frac{1}{25} = \frac{1}{5^2} = \frac{1 \times 2^2}{5^2 \times 2^2} = \frac{4}{100}$ b) $\frac{1}{25} = 0.04$	1 1	2
3	For identifying the largest angle in the triangle is 130^0 . Measure of each smaller angle $= \frac{180^0 - 130^0}{2} = 25^0$	1 1	2
4	Two times a number $= 21 - 5 = 16$ Number $= \frac{16}{2} = 8$ **OR** Number $= \frac{21 - 5}{2} = 8$	1 1	2
5	Area of the square $= \frac{1}{2} \times 10^2 = 50\ sq.cm$	2	2
	Each questions from 6 to 11 carries 3 scores. (Answer any 4)		
6	a) Quarterly rate of interest $= \frac{12}{4} = 3\%$ b) Amount get back after 6 months $= 6000\left(1 + \frac{3}{100}\right)^2$ $= 6000 \times \frac{103}{100} \times \frac{103}{100} = 6365.4$ Rs	1 1 1	3
7	a) Sum of parallel sides $= 8 + 4 = 12\ cm$ b) Area of the trapezium $= \frac{1}{2} \times 12 \times 5 = 30\ sq.cm$	1 2	3
8	a) $x^2 - y^2 = 10 \times 8 = 80$ b) $51 \times 49 = (50 + 1)(50 - 1)$ $= 50^2 - 1^2 = 2499$	1 1 1	3

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9

Weight (Kg)	Tally	Number of children
20 - 25	| |	2
25 - 30	𝍸	5
30 - 35	𝍸 | |	7
35 - 40	𝍸 | |	7
40 - 45	𝍸	5
45 - 50	| | | |	4

3 | 3

10

a) Number of triangles get when maximum number of diagonals can be drawn from a vertex of this polygon $= 10 - 2 = 8$ — 1

Sum of the inner angles $= 8 \times 180 = 1440^0$ — 1

b) Measure of an inner angle $= \frac{1440}{10} = 144^0$ — 1

3 | 3

11

a) $BD = 5 - 2 = 3\ cm$ — 1

b) Ratio between the areas of the triangles ADC and BDC $= AD : BD = 2 : 3$ — 1

c) Area of the triangle ADC $= \frac{2}{5} \times 35 = 14\ sq.cm$ — 1

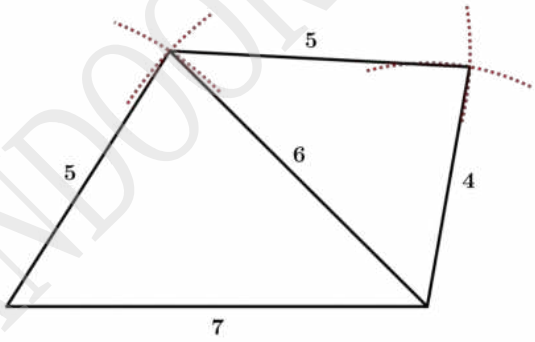
3 | 3

Each questions from 12 to 18 carries 4 scores. (Answer any 5)

12

For drawing a triangle with the 6 cm long diagonal as side . — 2

For drawing the second triangle with the 6 cm long diagonal as side and completing the quadrilateral. — 2



4 | 4

13

a) Sum of the angles $= 180^0$ — 1

b) For taking the measures of the angles as $2x^0$, $3x^0$ and $4x^0$ — 1

$2x + 3x + 4x = 180$ ==> $9x = 180$ ==> $x = \frac{180}{9} = 20$ — 1

Measures of the angles $= 2 \times 20$, 3×20 , $4 \times 20 = 40^0$, 60^0 , 80^0 — 1

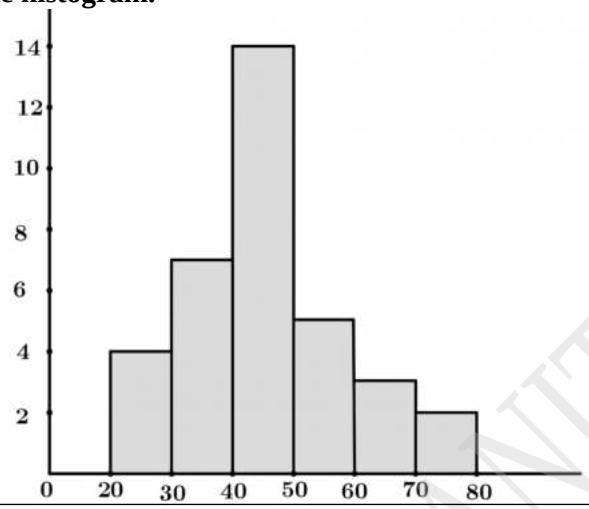
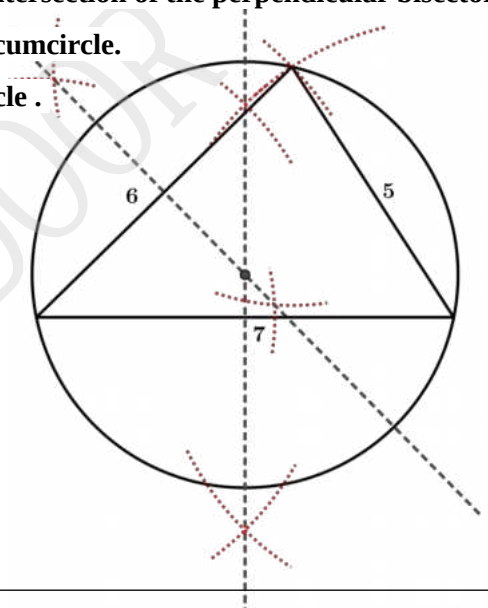
4 | 4

14

a) Length of a side $= \frac{40}{4} = 10\ cm$ — 1

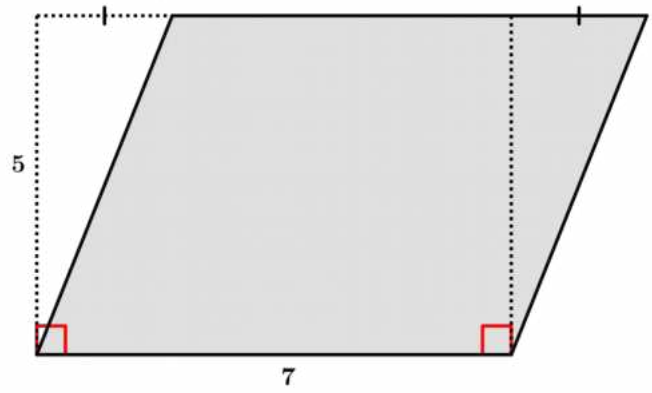
b) For identifying the diagonals of a rhombus bisect each other and perpendicular to reach other . — 1

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	Length of the second diagonal $= 2 \times \sqrt{10^2 - 6^2} = 2 \times 8 = 16\ cm$ **c) Area of the rhombus** $= \frac{1}{2} \times 12 \times 16 = 96\ sq.cm$	1 1	4
15	**For drawing horizontal and vertical lines .** **For marking classes on the horizontal line and frequencies on the vertical line.** **For completing the histogram.**  	1 1 2	4
16	**For drawing the triangle in the given measures.** **For drawing the perpendicular bisectors of the sides of the triangle.** **For identifying the point of intersection of the perpendicular bisectors of the sides of the triangle is the centre of its circumcircle.** **For completing the circumcircle .**  	1 1 1 1	4
17	**a) Area of the parallelogram** $= 6 \times 3 = 18\ sq.cm$ **b) For identifying the distance between the 7cm long sides is 5 cm .** **For completing the parallelogram** .	1 1 2	4

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18	**a)** $4^2 + 5^2 = 9^2 - 2(4 \times 5)$ **b)** 7×8 $\quad [\because 7^2 + 8^2 = 15^2 - 2(7 \times 8)]$ **c)** $20^2 + 21^2 = 41^2 - 2(20 \times 21)$ **d)** $a^2 + b^2 = (a+b)^2 - 2ab$	**1** **1** **1** **1**	**4**

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